

Mubsir AI

The Intelligent Shield for Digital Family Protection

A Comprehensive Multimodal AI Ecosystem for Proactive Threat Detection

PROJECT DETAILS

Graduation Project

AI for Business Track

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TECHNICAL & BUSINESS VISION DOCUMENT

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1. Executive Summary

"**Mubsir AI**" represents a quantum leap in family-oriented digital protection systems. It is not merely a traditional parental control application that relies on basic keyword blocking; rather, it is a comprehensive **Multimodal AI ecosystem** acting as a proactive shield to protect children from unethical content and dangerous digital behaviors.

The system is designed with an advanced engineering architecture that integrates **Computer Vision models** for real-time image analysis directly on edge devices (Edge Computing), alongside **Large Language Models (LLMs)**—represented by the **Gemini API**—to deeply analyze textual context and detect predatory grooming patterns.

This project aims to bridge a critical gap in both the B2C and B2B (AI for Business) markets by offering a scalable product that ensures absolute user privacy while delivering an innovative technical solution to a pressing societal issue.

2. Problem & Vision

2.1 Current Market Challenges

- **Easy Evasion:** Most current applications rely on static keyword blacklists that tech-savvy children can easily bypass.
- **Visual & Spatial Shortcomings:** Existing apps often fail to understand context, leading to high False Positive rates (e.g., inappropriately blocking safe, natural images).
- **Ignoring Psychological Danger:** Traditional focus is solely on images, whereas the most significant threats often lie in textual conversations, such as cyberbullying and grooming.

2.2 The Innovation in "Mubsir AI"

The system introduces a solution grounded in "**Contextual Understanding**". Instead of examining an image in isolation, the system detects the presence of people and identifies their gender and age group. Simultaneously, the Gemini API reads between the lines in conversations to uncover any psychological manipulation or threats directed at the child.

3. AI Architecture

To guarantee the highest standards of accuracy and speed, we did not rely on a monolithic model. Instead, we constructed an intelligent pipeline that fuses three specialized vision models supported by a robust linguistic engine.

3.1 Ensemble Vision Models

Model	Architecture	Size	Functional Role
NSFW Classifier	MobileNetV2 (TFLite)	22 MB	Primary content classification into 5 distinct categories with an accuracy of 96% .
BlazeFace	TFLite	224 KB	Real-time human face detection to establish and confirm context.
FaceRes	MobileNet (TF.js)	6.8 MB	Extraction of demographic data (Gender and Age estimation).

Table 1 — The constituent models of Mubsir AI's vision engine

⚡ Custom Primary Model (MobileNetV2)

This model is the beating heart of the visual system:

- **Training Data:** Trained on a massive, complex dataset containing **130,000 images** (collected via deep scraping from multiple sources).
- **Accuracy & Performance:** Leveraging the MobileNetV2 architecture, the model achieved up to **96% accuracy** and was optimized into a TFLite format to run efficiently on Android devices without internet connectivity.
- **Anti-SFW Bias Correction:** An engineering innovation to minimize false negatives; probabilities for safe classifications are multiplied by a coefficient (0.6). This dynamically heightens the system's sensitivity to dangerous content.

3.2 LLM NLP Engine

To complete the defensive shield, the **Gemini API** was integrated as an Intelligent Agent operating seamlessly in the background:

- **Intent Analysis:** Transcends single-word filtering to understand the true intent of the sender in chats and social media applications.
- **Proactive Alerting:** Monitors and identifies early signs of cyberbullying, extortion, or grooming attempts before they can escalate into physical or psychological harm.

4. Engineering Challenges

Building a system of this magnitude required overcoming complex technical hurdles:

- **Data Sourcing Challenge:** Obtaining 130,000 accurately classified images was a significant undertaking. *Solution:* Deep scraping operations were executed across GitHub and Kaggle repositories, followed by extensive merging, cleansing, and curation to build a robust proprietary database.
- **Data Imbalance Challenge:** Certain content categories overwhelmingly dominated the dataset, posing a severe risk of model bias. *Solution:* Programmatic intervention was applied using intensive Data Augmentation techniques, alongside the meticulous fine-tuning of optimal decision thresholds for each individual class.
- **Cloud Deployment Challenge:** Deploying deep learning models to edge and cloud environments often hits severe memory and size limits. *Solution:* Transitioning from the full TensorFlow framework to the highly optimized `tflite`-runtime, drastically reducing the system footprint and enabling instant, error-free deployments on platforms like Streamlit Cloud.

5. The Product

This advanced architecture culminated in a comprehensive suite comprising a fully-fledged **Android Application** and a robust **Streamlit Web Interface**.

The mobile application boasts the capability to run silently in the background, safeguarding the user through **Smart Auto-blur** technology. It applies instantaneous blurring filters to inappropriate visual content before it even reaches the child's eyes. Simultaneously, the **Streamlit Web Interface** allows for real-time model evaluation, visual

probability tracking, and scalable cloud deployment, making the ecosystem universally accessible.

6. Business Viability

"Mubsir AI" perfectly aligns with the core objectives of the **AI for Business** track:

1. **Privacy First (Edge AI):** Executing the vision models locally on the user's smartphone (via TFLite) ensures that no personal images are ever uploaded to the cloud, establishing absolute trust with the end-user (B2C).
2. **Cost Efficiency:** Processing images locally eliminates the need for heavy server-side processing, saving thousands of dollars in monthly cloud infrastructure costs.
3. **Scalability (B2B SaaS):** The underlying architecture can easily be packaged as an API for telecommunications companies and Internet Service Providers (ISPs) to integrate directly into home routers, unlocking massive investment and enterprise horizons.